

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

Considering the major vulnerabilities explained, it's necessary to implement MFA and password policies. MFA is not used and the organization has poor password policies, the employees share them and some of the passwords, like the database password, has the default value.

MFA requires users to use more than one way to identify and verify their credentials before accessing an application. Some MFA methods include fingerprint scans, ID cards, pin numbers, and passwords.

Password policies can be refined to include rules regarding password length, a list of acceptable characters, and a disclaimer to discourage password sharing. They can also include rules surrounding unsuccessful login attempts, such as the user losing access to the network after five unsuccessful attempts.

Part 2: Explain your recommendations

These techniques are effective because they address 3 of the major vulnerabilities described:

1. The organization's employees' share passwords.
2. The admin password for the database is set to the default.
3. Multifactor authentication (MFA) is not used.

Enforcing multi-factor authentication (MFA) adds an additional layer of security beyond a password. It will reduce the likelihood that a malicious actor can access a network through a brute force or related attack since additional effort is required to authenticate in more than one way.

Creating and enforcing a password policy within the company will make it increasingly challenging for malicious actors to access the network. Policies such as suspending the account after a certain number of logins can prevent

successful brute force attacks. Increasing password complexity, requiring more frequent password updates, and not allowing passwords to be reused also help stall malicious actors from infiltrating the network.

Password policies and MFA should be implemented once, and then they should be monitored to determine if they are effectively being used or working effectively.